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1. INTRODUCTION

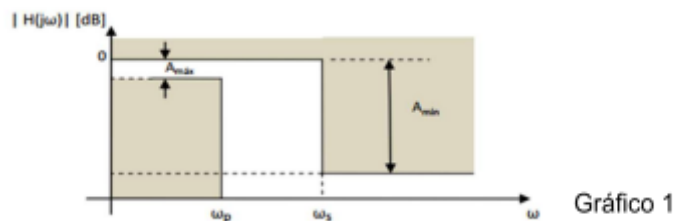
By using operational amplifiers in conjunction with resistors and capacitors, it is possible to create filters to modify various signals without the need to add inductors. Over the past few weeks, we have been able to understand several methods for obtaining a low-pass filter transfer function that satisfies the imposed conditions. Using this as a prototype, and applying frequency transformation concepts, it is possible to obtain different types of filters that approximate the behavior of band-pass, band-stop, or high-pass filters.

1.1 Objectives

- Identify the main characteristics of approximation functions
- Design active filters that allow compliance with the specified requirements.
- Handle circuit simulation and numerical tools.
- Implement the design and take measurements on it

1.2 Conditions

Based on Figure 1, design a band-stop filter with maximum flatness, meeting a passband of $f < 3.6\text{kHz}$ and $f > 9.1\text{kHz}$ and a stopband of $5.45\text{kHz} < f < 5.9\text{kHz}$. It must also meet a minimum attenuation of 38dB and a maximum attenuation of 1.5dB.



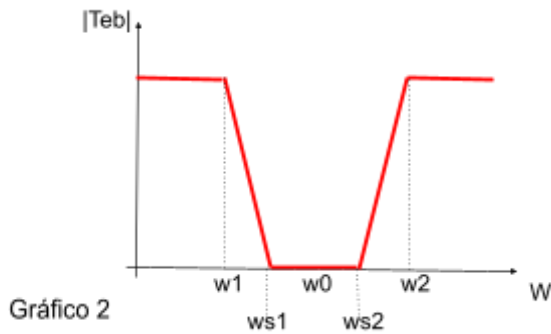
General template for filters.

Its implementation must be through the second-order Ackerberg-Mossberg structure.

2. DEVELOPMENT

2.1 Low-pass calculations

Based on the indicated conditions, we created the template corresponding to the attenuation and gain (graph 2 and 3) with the frequencies of the passband and rejection band.



Band-eliminating filter transfer.

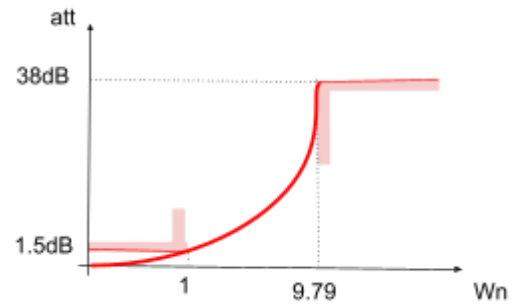


Gráfico 3

Filter template based on attenuation.

Firstly, it is necessary to obtain IN_0 and Q through:

$$IN_0 = \sqrt{IN_1 * IN_2} = \sqrt{(2\pi * 3k6Hz) * (2\pi * 9k1Hz)} = 35962,7 rad/s$$

$$Q = \frac{IN_0}{IN_2 - IN_1} = \frac{35962,7}{18200\pi - 7200\pi} = 1.04$$

To be able to work correctly, we normalize all the data with respect to IN_0

$$IN_{0N} = 1 \quad IN_{1N} = 0,63 \quad IN_{2N} = 1,59 \quad IN_{S1N} = 0,95 \quad IN_{S2N} = 1,03$$

By analyzing the conditions of the stop frequencies and their corresponding attenuations, we see that $IN_1 * IN_2 \simeq 1$ and $to_{min1} = to_{min2}$ and consequently we can calculate them all with a single value for the passband edge frequency Ω_s .

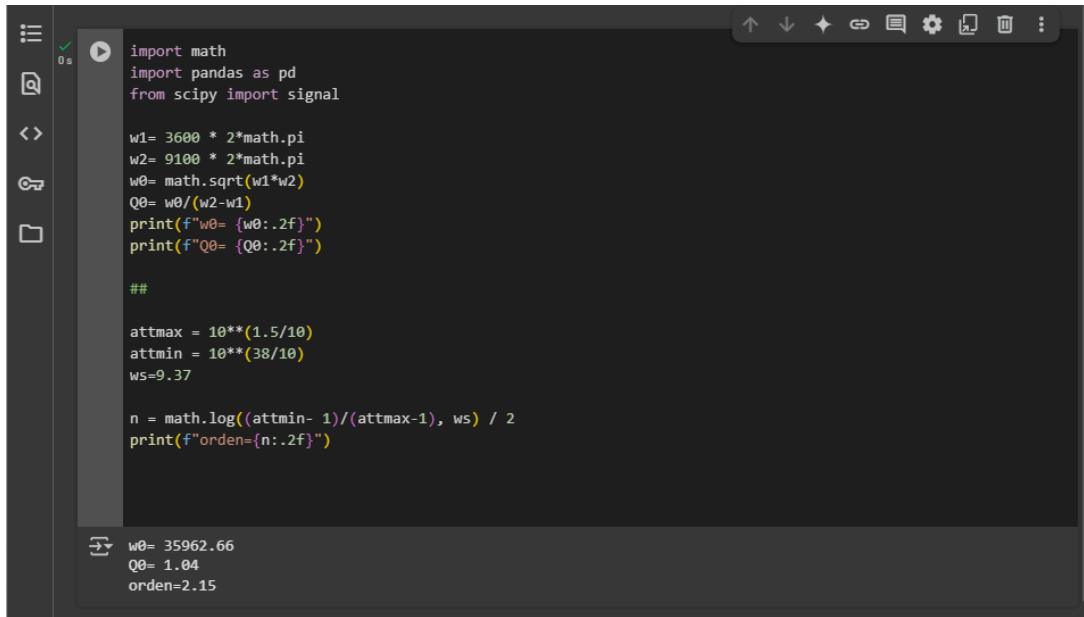
$$\Omega_{s1} = -\frac{In_s/Q}{In_s^2 - 1} = -\frac{0,95/1,04}{0,95^2 - 1} = 9,37$$

$$\Omega_{s2} = -\frac{In_s/Q}{In_s^2 - 1} = -\frac{1,03/1,04}{1,03^2 - 1} = -16,26$$

With the calculated data we can obtain the filter order for maximum flatness and the ξ scaled to our conditions.

$$n = \frac{\log[(10^{0,1 * \varphi_{min}} - 1) / (10^{0,1 * \varphi_{max}} - 1)]}{2 * \log(In_s)} = 2,15$$

Figure 4 shows the verification using Python.



```

import math
import pandas as pd
from scipy import signal

w1= 3600 * 2*math.pi
w2= 9100 * 2*math.pi
w0= math.sqrt(w1*w2)
Q0= w0/(w2-w1)
print(f"w0= {w0:.2f}")
print(f"Q0= {Q0:.2f}")

##

attmax = 10**(1.5/10)
attmin = 10**(38/10)
ws=9.37

n = math.log((attmin- 1)/(attmax-1), ws) / 2
print(f"orden={n:.2f}")

```

w0= 35962.66
Q0= 1.04
orden=2.15

Gráfico 4

Calculating the values using Python.

$$\xi = \sqrt{10^{0,1\varphi_{max}} - 1} = \sqrt{10^{0,1,1,5} - 1} = 0,6423$$

$$\frac{\$}{\sqrt{\xi}} = \frac{\$}{0,8014}$$

At this point, the correct decision is to round up and take $n=3$, but considering the difference in decimals and the simplicity that using order 2 could bring when designing, we compare both cases.

$$T_{PB(n=2)}(\$) = \frac{1}{\$^2 + \$\sqrt{2} + 1}$$

$$T_{PB(n=3)}(\$) = \frac{1}{\$^2 + \$ + 1} * \frac{1}{\$ + 1}$$

2.2 Band-eliminating adaptation

Once the necessary data was obtained by reading the template and performing additional calculations, we implemented the corresponding equation to find the transfer of a band-stop filter.

$$\$ = \frac{1}{Q} * \frac{1}{s + \frac{1}{s}} = \frac{1}{1,04} * \frac{1}{s + \frac{1}{s}} = \frac{0,9615}{s + \frac{1}{s}}$$

We must not forget the adaptation made earlier for the 1.5dB attenuation.

$$\frac{\$}{\sqrt{\xi}} = \frac{0,9615}{s + \frac{1}{s}} = \frac{\$}{0,8014} \Rightarrow \$ = \frac{0,7705}{s + \frac{1}{s}}$$

It would simply be necessary to substitute this new expression into the corresponding transfer function, both for n=2 and for n=3, in order to be able to compare.

n=2

$$\begin{aligned} T_{EB}(s) &= \frac{1}{\left(\frac{0,7705}{s + \frac{1}{s}}\right)^2 + \sqrt{2} \left(\frac{0,7705}{s + \frac{1}{s}}\right) + 1} = \frac{1}{\frac{0,59s^2}{(s^2+1)^2} + \frac{\sqrt{2} \cdot 0,7705 \cdot s + (s^2+1)}{s^2+1}} = \frac{1}{\frac{0,59s^2 + [\sqrt{2} \cdot 0,7705 \cdot s + (s^2+1)](s^2+1)}{(s^2+1)^2}} \\ &= \frac{(s^2+1)^2}{0,59s^2 + 1,088s^3 + 1,08s + s^4 + 2s^2 + 1} = \frac{(s^2+1)^2}{s^4 + s^3 \cdot 1,08 + s^2 \cdot 2,59 + s \cdot 1,08 + 1} = \frac{s^4 + 2s^2 + 1}{(s^2 + s \cdot 0,69 + 1,75)(s^2 + s \cdot 0,396 + 0,573)} \\ &= \frac{s^2+1}{s^2 + s \cdot 0,69 + 1,75} * \frac{s^2+1}{s^2 + s \cdot 0,396 + 0,573} \\ &= \frac{(s^2+1)^2}{[s - (-0,346 - 1,27j)] * [s - (-0,346 + 1,27j)] * [s - (0,198 + 0,73j)] * [s - (-0,198 - 0,73j)]} \end{aligned}$$

n=3

$$\begin{aligned} T_{EB}(s) &= \frac{1}{\left(\frac{0,7705}{s + \frac{1}{s}}\right)^2 + \left(\frac{0,7705}{s + \frac{1}{s}}\right) + 1} * \frac{1}{\left(\frac{0,7705}{s + \frac{1}{s}}\right) + 1} = \frac{(s^2+1)^2}{(s^2 + s \cdot 0,769 + 1)(s^4 + s^3 \cdot 0,769 + s^2 \cdot 2 + s \cdot 1,36 + 1)} \\ &= \frac{(s^2+1)^2}{(s^2 + s \cdot 0,769 + 1)(s^2 + s \cdot 0,966 + 0,649)(s^2 + s \cdot 1,196s + 1,54)} \\ &= \frac{(s^2+1)^2}{[s - (0,385 + 0,927j)] * [s - (0,385 - 0,927j)] * [s - (-0,483 + 0,645j)] * [s - (-0,483 - 0,645j)] * [s - (0,098 + 1,237j)] * [s - (0,098 - 1,237j)]} \end{aligned}$$

From this point on, only the order n=2 will be used. In graph 5, we present the pole-zero diagram corresponding to the band-stop filter transfer.

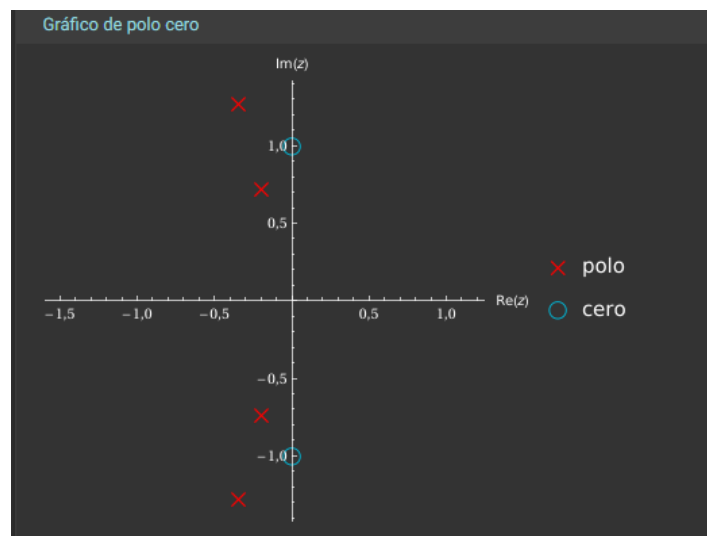


Gráfico 5

Poles and zeros of $T_{eb}(n=4)$.

We also present the expected filter frequency response in Figure 6; where the dotted lines represent the cutoff (red) and passband (blue) frequencies.

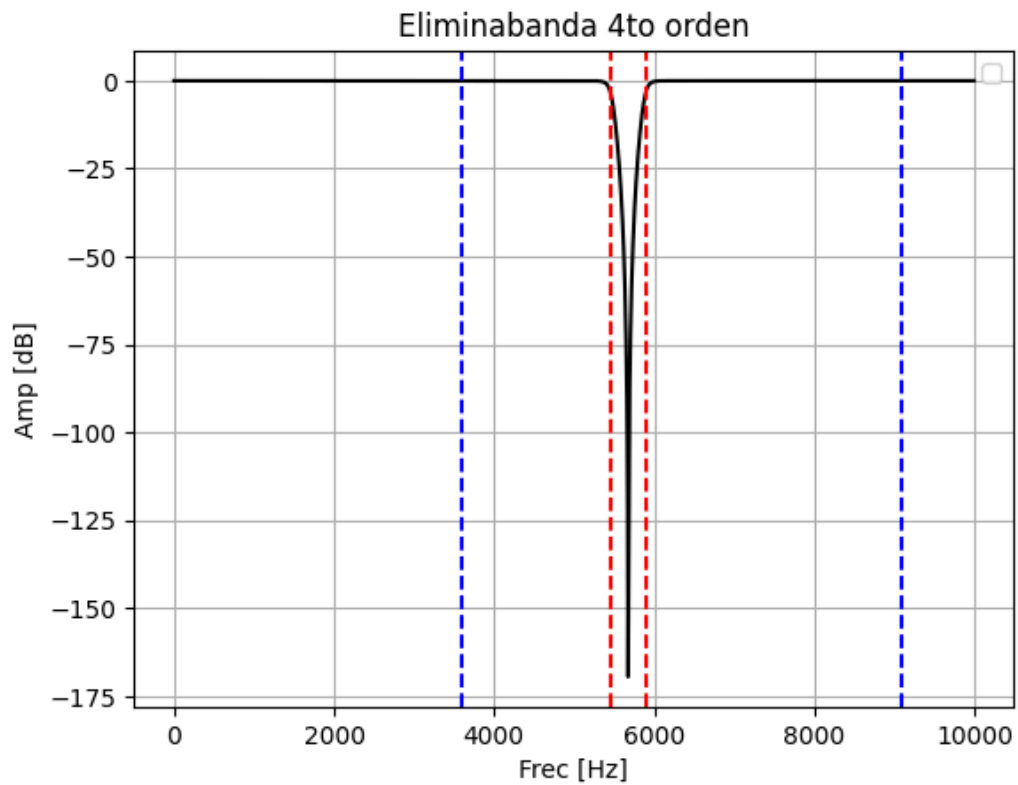


Gráfico 6

Frequency response of the band-stop filter with its pass and cutoff frequencies.

2.3 Module and phase

It is necessary to obtain the expressions that represent the magnitude, phase, and delay by transferring the filter as a function of omega. The corresponding graphical representations can be seen in Figures 7 and 8.

$$T_{EB}(jw) = \frac{(jw)^4 + 2(jw)^2 + 1}{In^4 - jIn^3 1,08 - 2,59In^2 + jw 1,08 + 1} = \frac{In^4 - 2In^2 + 1}{(In^4 - 2,59In^2 + 1) + j(-1,08w^3 + 1,08w)}$$

→ Module

$$|T_{EB}(jw)| = \frac{|In^4 - 2In^2 + 1|}{|(In^4 - 2,59In^2 + 1) + j(-1,08w^3 + 1,08w)|}$$

$$= \frac{In^4 - 2In^2 + 1}{\sqrt{(In^4 - 2,59In^2 + 1)^2 + (-1,08w^3 + 1,08w)^2}}$$

→ Phase

$$\phi[T_{EB}(jw)] = \arctg\left[\frac{0}{(In^4 - 2,59In^2 + 1)}\right] - \arctg\left[\frac{(-1,08w^3 + 1,08w)}{(In^4 - 2,59In^2 + 1)}\right]$$

$$= -\arctg\left[\frac{(-1,08w^3 + 1,08w)}{(In^4 - 2,59In^2 + 1)}\right]$$

→ Delay

$$D[T_{EB}(in)] = -\frac{d[T_{EB}(in)]}{dw} = \frac{1,08In^6 - 0,44In^4 - 0,44In^2 + 1,08}{In^8 - 4In^6 + 6,35In^4 - 4In^2 + 1}$$

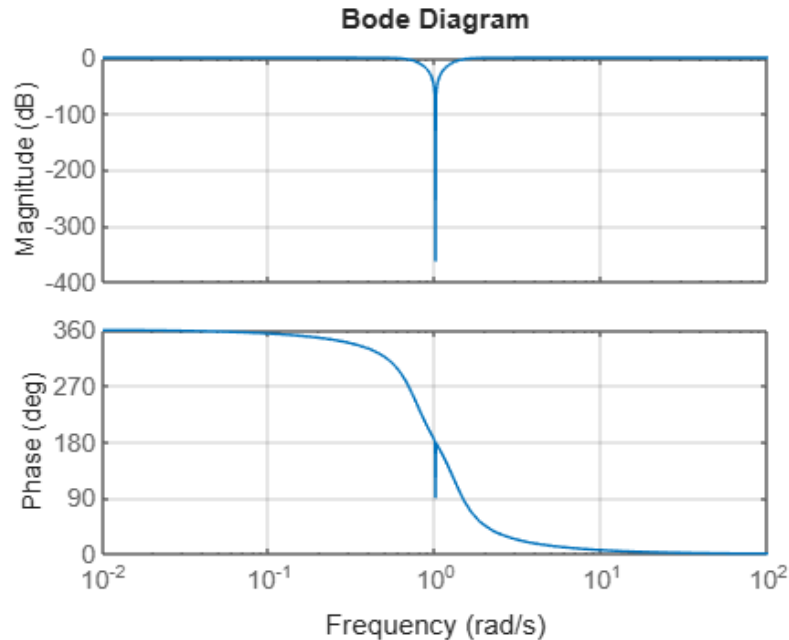


Gráfico 7

Graph of the module and phase from Matlab.

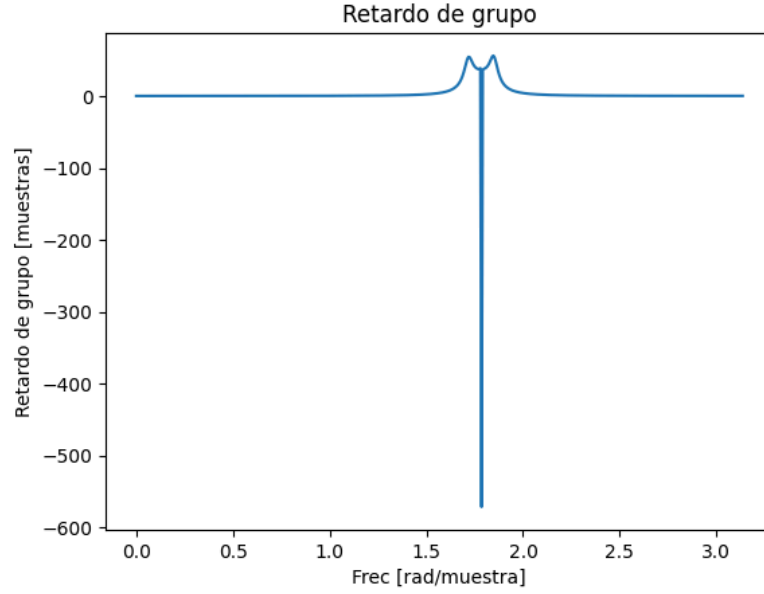


Gráfico 8

“Delay Group” of the created filter.

2.4 Arquitectura Ackerberg

For the design corresponding to the two biquadratic functions obtained for order 2, we will use the Ackerber-Mossberg circuit of four operational amplifiers with the corresponding resistors as shown in graph 9.

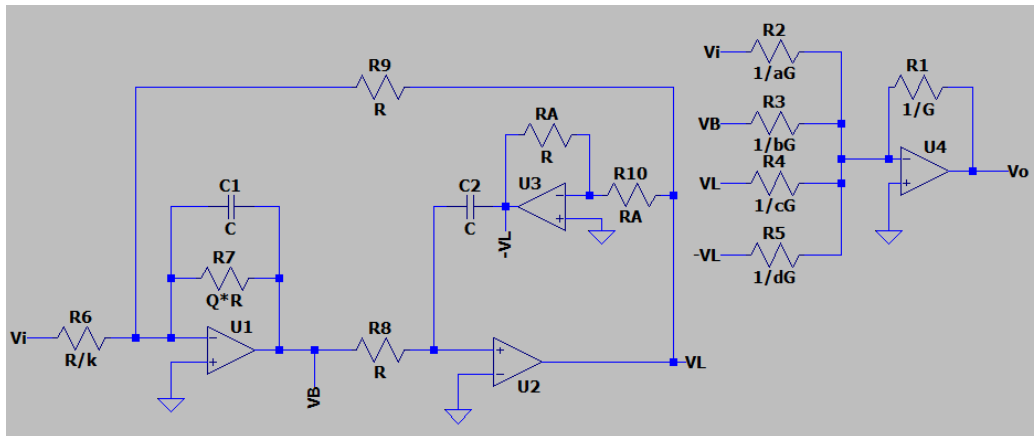


Gráfico 9

Estructura Ackerberg-Mosberg.

$$H(s) = \frac{a s^2 + \frac{\text{Where}}{Q} (a - Q^* k^* b) + W t h e^2 (a - (c - d)^* k)}{s^2 + \frac{\text{Where}}{Q} s + W t h e^2}$$

It is necessary to know the quality factors of each of the quadratic blocks

$$H(s) = \frac{(s^2+1)*(s^2+1)}{(s^2 + s 0,69 + 1,75)*(s^2 + s 0,396 + 0,573)} = \frac{(s^2+1)}{(s^2 + s 0,69 + 1,75)} * \frac{(s^2+1)}{(s^2 + s 0,396 + 0,573)}$$

$$IN_1^2 = 1,75 \quad \quad \quad IN_2^2 = 0,573$$

$$\frac{IN_1}{Q_1} = 0.69 \text{ with } \Rightarrow Q_1 = 1,92$$

$$\frac{IN_2}{Q_2} = 0.396 \text{ with } \Rightarrow Q_2 = 1,91$$

Finally, we combine the information obtained with the boundary conditions of a band-removal filter to obtain the values we will work with.

$$\text{I) } a = 1$$

$$\text{II) } INthe^2(a - (c - d)k) = 1 \quad (\text{con } k=1)$$

$$(a - (c - d)) = \frac{1}{INthe^2}$$

$$1 - \frac{1}{INthe^2} = c - d$$

$$(c - d)_1 = 1 - \frac{1}{IN_1^2} = 0,429$$

$$c_1 = 0,429 \quad \& \quad d_1 = 0$$

$$(c - d)_2 = 1 - \frac{1}{IN_2^2} = -0,745$$

$$c_2 = 0 \quad \& \quad d_2 = 0,745$$

$$\text{III) } a - Qkb = 0 \quad (\text{con } k=1)$$

$$a = Qkb$$

$$b = \frac{1}{Qk}$$

$$b = \frac{1}{Q}$$

$$b = 0,524$$

It is also necessary to obtain the standardized resistances of the system.

With $C_{1N} = C_{2N} = 1$ and taking into account that

$$IN_1 = \frac{1}{R^*C} = 1,32 \quad \text{and} \quad IN_2 = \frac{1}{R^*C} = 0,757$$

$$\bullet \quad R_{1N} = \frac{1}{C_{1N} * IN_1} = 0,756$$

$$\bullet \quad R_{2N} = \frac{1}{C_{2N} * IN_2} = 1,321$$

In this way, we can now obtain the actual values of the components for each part of the complete circuit. To facilitate obtaining these values when creating the circuit board, we impose the capacitor values for two cases.

$$1) C_1 = C_1 = 100nF$$

$$Oh_N = 35962,7 \quad WITH_N = \frac{C_N}{C * Oh_N} = 278,07$$

- $R_1 = R_{1N} * WITH_N = 0.756 * 278.07 = 210.22\Omega$
- $R_2 = R_{2N} * WITH_N = 1.321 * 278.07 = 367.33\Omega$
- $Q * R_1 = 1.91 * 210.22\Omega = 401.52\Omega$
- $Q * R_2 = 1.91 * 367.33\Omega = 701.6\Omega$

$$2) C_1 = C_1 = 10nF$$

$$Oh_N = 35962,7 \quad WITH_N = \frac{C_N}{C * Oh_N} = 2780,66$$

- $R_1 = R_{1N} * WITH_N = 0.756 * 2780.66 = 2102.18\Omega$
- $R_2 = R_{2N} * WITH_N = 1.321 * 2780.66 = 3673.25\Omega$
- $Q * R_1 = 1.91 * 2102.18\Omega = 4015.16\Omega$
- $Q * R_2 = 1.91 * 3673.25\Omega = 7015.9\Omega$

In both cases, we chose $RA = 2k2\Omega$ and $\frac{1}{G} = 1k\Omega$

- $\frac{1}{aG} = \frac{1}{G} = 1k\Omega$
- $\frac{1}{bG} = \frac{1}{0,524} * 1k\Omega = 1908.4\Omega$
- $\frac{1}{c_1G} = \frac{1}{0,429} * 1k\Omega = 2331\Omega$
- $\frac{1}{c_2G} = \frac{1}{0,0} * 1k\Omega \Rightarrow \infty$
- $\frac{1}{d_1G} = \frac{1}{0,0} * 1k\Omega \Rightarrow \infty$
- $\frac{1}{d_2G} = \frac{1}{0,745} * 1k\Omega = 1342.28\Omega$

Based on the calculations, we look for the closest commercial values to avoid a greater error between the theoretical and the practical, while maintaining what was imposed for SUN and $\frac{1}{G}$.

$$1) C_1 = C_1 = 100nF$$

- $R_1 = 220\Omega$
- $R_2 = 390\Omega$
- $\frac{1}{aG} = \frac{1}{G} = 1k\Omega$
- $\frac{1}{bG} = 1k8\Omega$
- $\frac{1}{c_1G} = 2k2\Omega$
- $\frac{1}{d_2G} = 1k2\Omega$
- $Q * R_1 = 390\Omega$
- $Q * R_2 = 680\Omega$

$$2) C_1 = C_1 = 10nF$$

- $R_1 = 2200\Omega$
- $R_2 = 3900\Omega$
- $\frac{1}{aG} = \frac{1}{G} = 1k\Omega$
- $\frac{1}{bG} = 1k8\Omega$
- $\frac{1}{c_1G} = 2k2\Omega$
- $\frac{1}{d_2G} = 1k2\Omega$
- $Q * R_1 = 3900\Omega$
- $Q * R_2 = 6800\Omega$

3. SIMULATION

3.1 Paper securities

Based on the circuit in Figure 10, we present the simulation with previously calculated values of the band-stop filter using the capacitors of $100nF$ Output signal in graph 11 and cursors for the frequencies of interest in graphs 12 and 13.

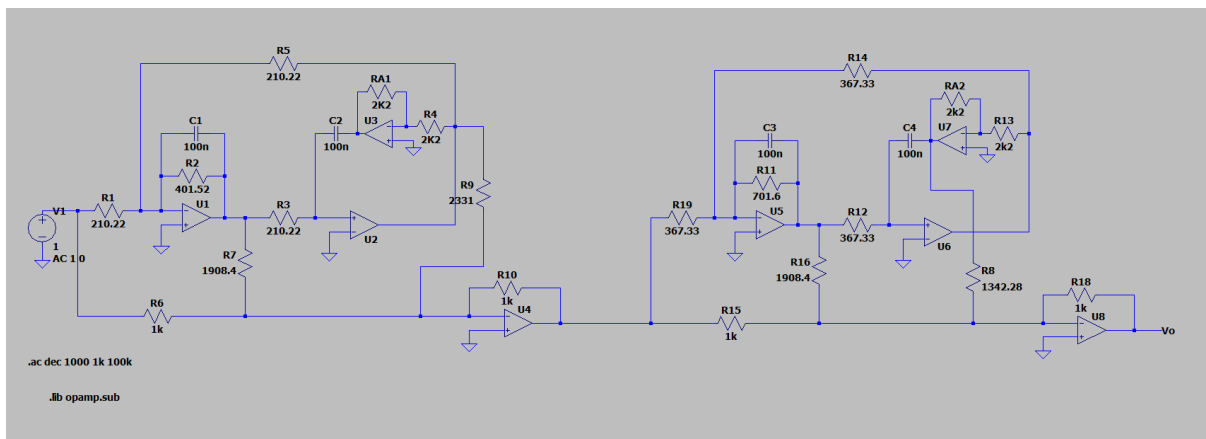


Gráfico 10

band-removal filter with calculated values.

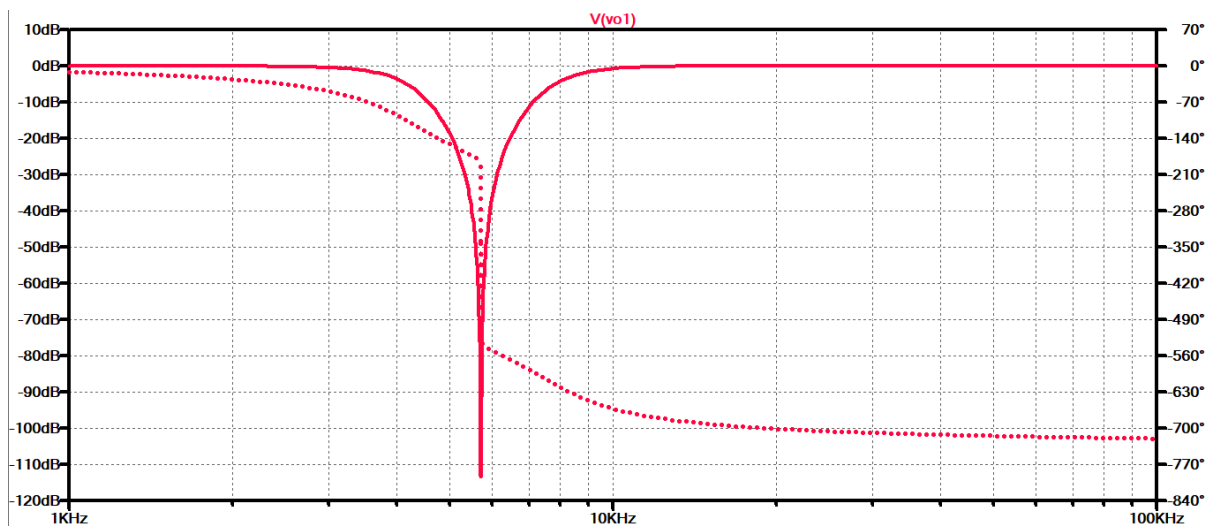


Gráfico 11

Filter response with calculated values.

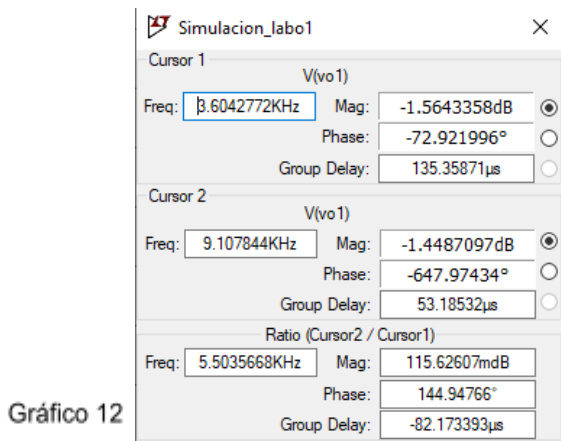


Gráfico 12

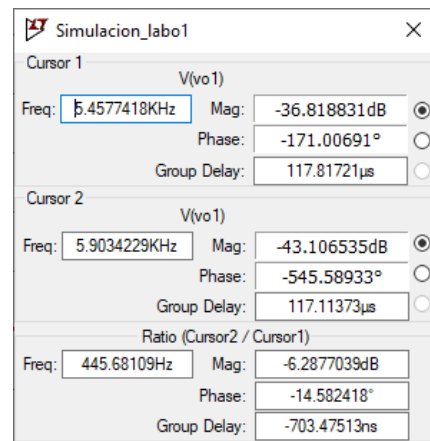


Gráfico 13

Cursors for the passband and cutoff frequencies.

3.2 Adaptation to commercial

In a second instance, we replaced the theoretical values with the most appropriate commercial ones (graph 14).

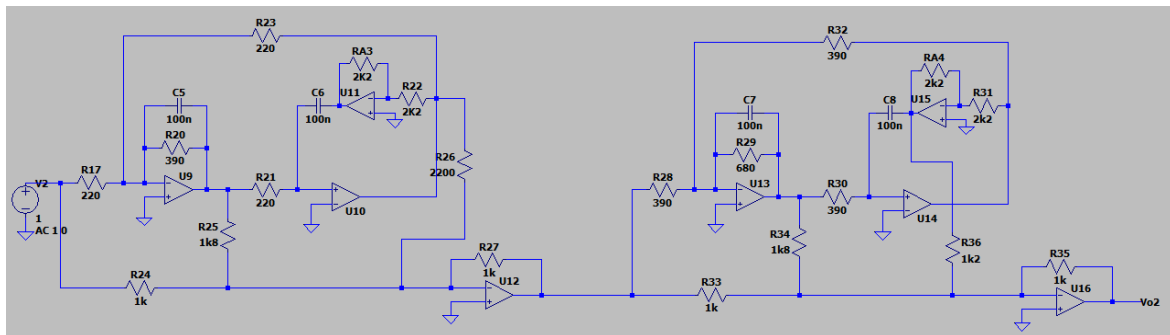


Gráfico 14

band-eliminator filter with commercial values.

Graph 15 shows the response obtained, while graphs 16 and 17 again present the values according to the cursors.

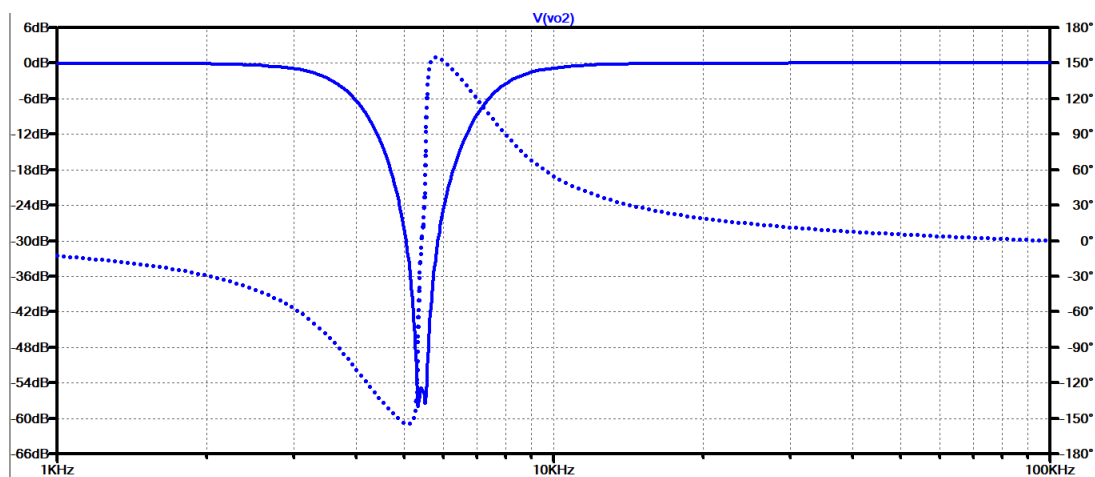


Gráfico 15

Filter response with commercial values.

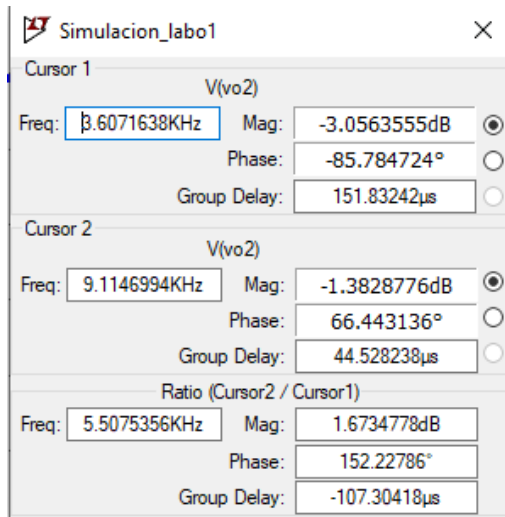


Gráfico 16

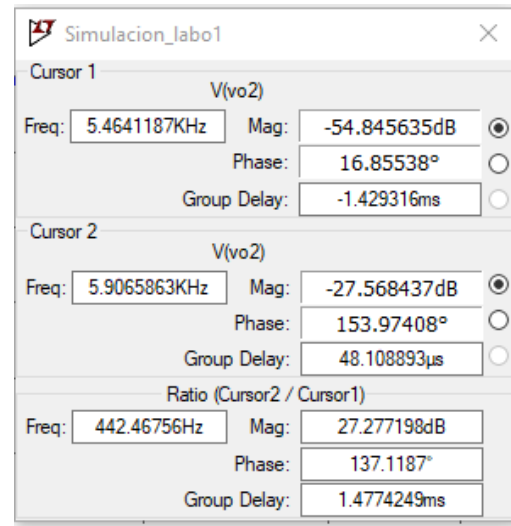


Gráfico 17

Cursors for the passband and cutoff frequencies.

In order to analyze the differences between both cases, it is useful to redraw the pole-zero diagram with its corresponding transfer.

$$H(s) = \frac{a s^2 + \frac{Where}{Q}(a - Q * k * b) + W t h e^2(a - (c - d) * k)}{s^2 + \frac{Where}{Q}s + W t h e^2}$$

Where the $Where_1$ Calculated using standardized commercial components, we end

$$\text{up with: } Where_1 = \frac{1}{R_{1N} * C_{1N}} = \frac{1}{0,7910h * 1F} = 1,264 \frac{1}{s}$$

It should be noted that R_{1N} worth 0.791Ω since it was cleared from the following equation:

$$R_{1N} = \frac{R_1}{W I T H_N} = \frac{2200h}{278,07} = 0.791\Omega$$

Then, he $Where_2$ Calculated using standardized commercial components, we end up

$$\text{with: } Where_2 = \frac{1}{R_{2N} * C_{2N}} = \frac{1}{1,4020h * 1F} = 0,713 \frac{1}{s}$$

It should be noted that R_{2N} worth $1,402\Omega$ since it was cleared from the following equation:

$$R_{2N} = \frac{R_2}{W I T H_N} = \frac{3900h}{278,07} = 1,402\Omega$$

Continuing, we calculate with the new values of $Where_1$ and $Where_2$ a $H_1(s)$ y $H_2(s)$ respectively.

$$H_1(s) = \frac{1 * s^2 + \frac{1,264}{1,91}(1 - 1,91 * 1 * 0,524) + 1,264^2(1 - (0,424 - 0) * 1)}{s^2 + \frac{1,264}{1,91}s + 1,264^2}$$

$$H_1(s) = \frac{s^2 + 0,92}{s^2 + 0,661s + 1,597}$$

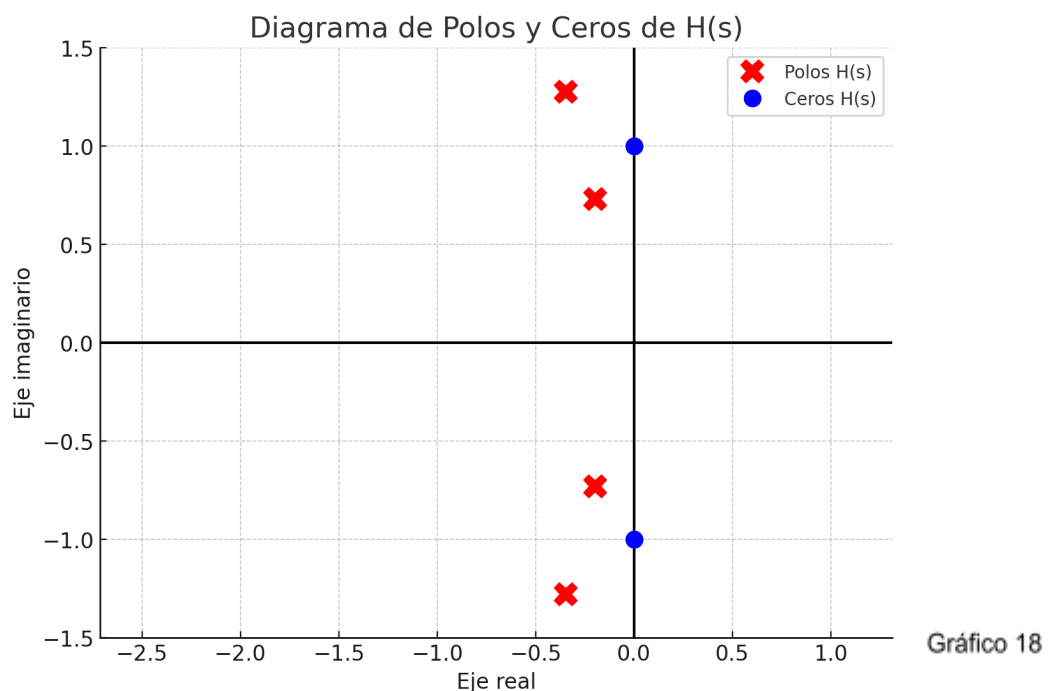
$$H_2(s) = \frac{1 * s^2 + \frac{0,713}{1,91}(1 - 1,91 * 1 * 0,524) + 0,713^2(1 - (0 - 0,745) * 1)}{s^2 + \frac{0,713}{1,91}s + 0,713^2}$$

$$H_2(s) = \frac{s^2 + 0,887}{s^2 + 0,373s + 0,508}$$

It can be seen that the new transfers still exhibit the behavior of a band-stop filter. However, both the poles and zeros have now shifted slightly due to the use of commercial components. Most notably, unlike the ideal transfers before, the zeros no longer occupy the same position on the s plane.

Next, in graphs 18 and 19, the pole-zero diagrams of the ideal transfers and the transfers with commercial components will be presented in order to see graphically how the locations of the poles and zeros have been modified.

Ideal transfer:



Poles and zeros of ideal shape.

-There is a double zero in: $s = \pm j$

-The poles are located in: $s = -0,345 \pm 1,222j$ and $s = -0,198 \pm 0.706j$

Transfer with commercial components:

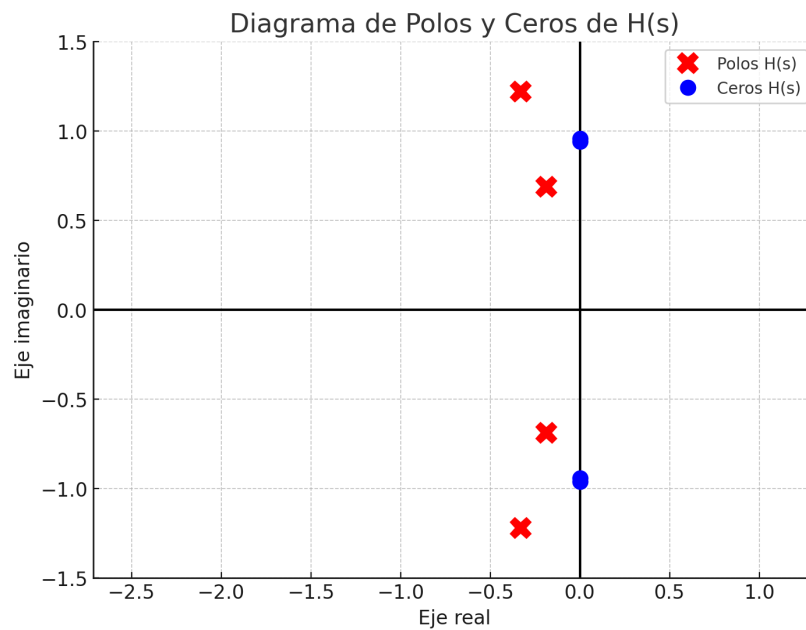


Gráfico 19

Poles and zeros with commercial values.

- The zeros are located in: $s = \pm 0.96j$ and in $s = \pm 0.94j$
- The poles are located in: $s = -0.33 \pm 1.15j$ and $s = -0.187 \pm 0.67j$

It is concluded that, although the difference is minimal, the zeros do not coincide in the same position as in the ideal case. And that both the zeros and the poles have decreased their distance with respect to the origin of coordinates.

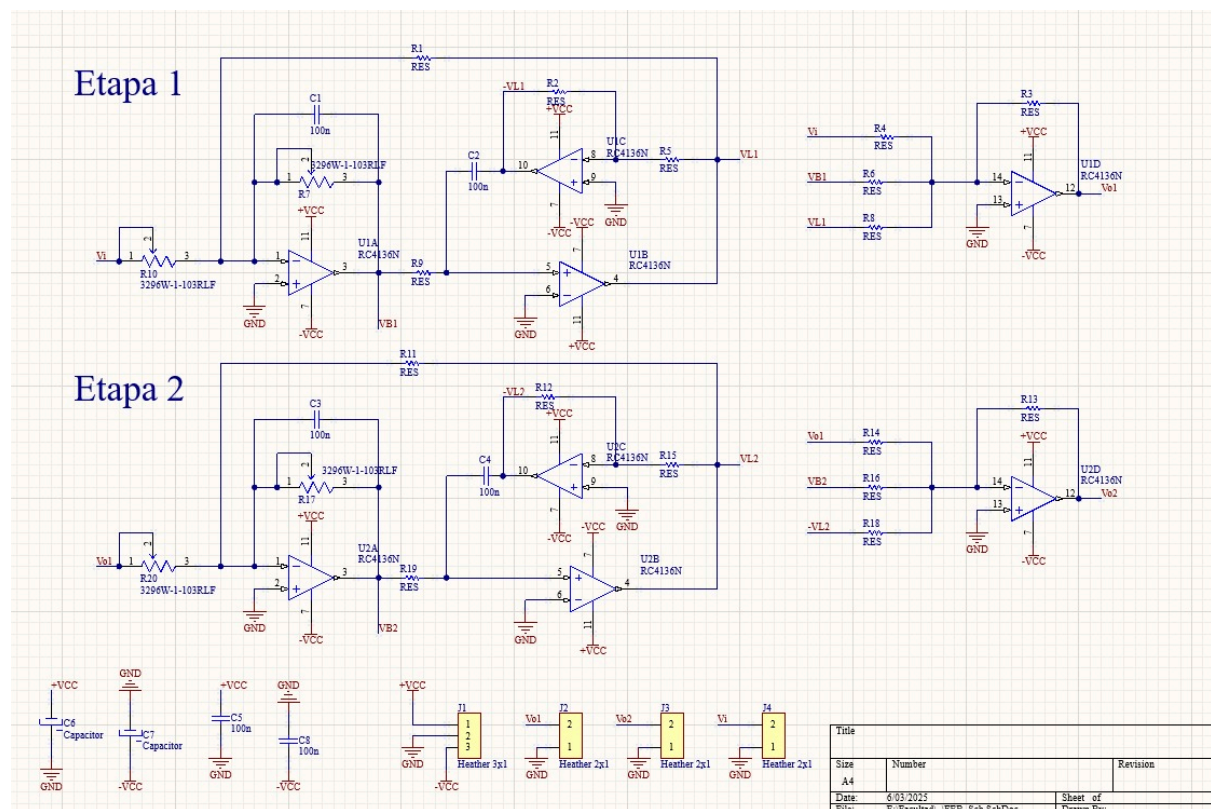
4. PLATE DESIGN

Initially, we used "Altium Designer" to implement the circuit on a physical board. To facilitate connecting the traces from the design stage, we decided to create a double-sided board; however, this—along with some issues during the mirroring process—led to several problems when soldering the components.

Because of this, a second version was designed in “KiCad” in a simplified way, with a single layer and with easier-to-solder tracks.

4.1 Schematic

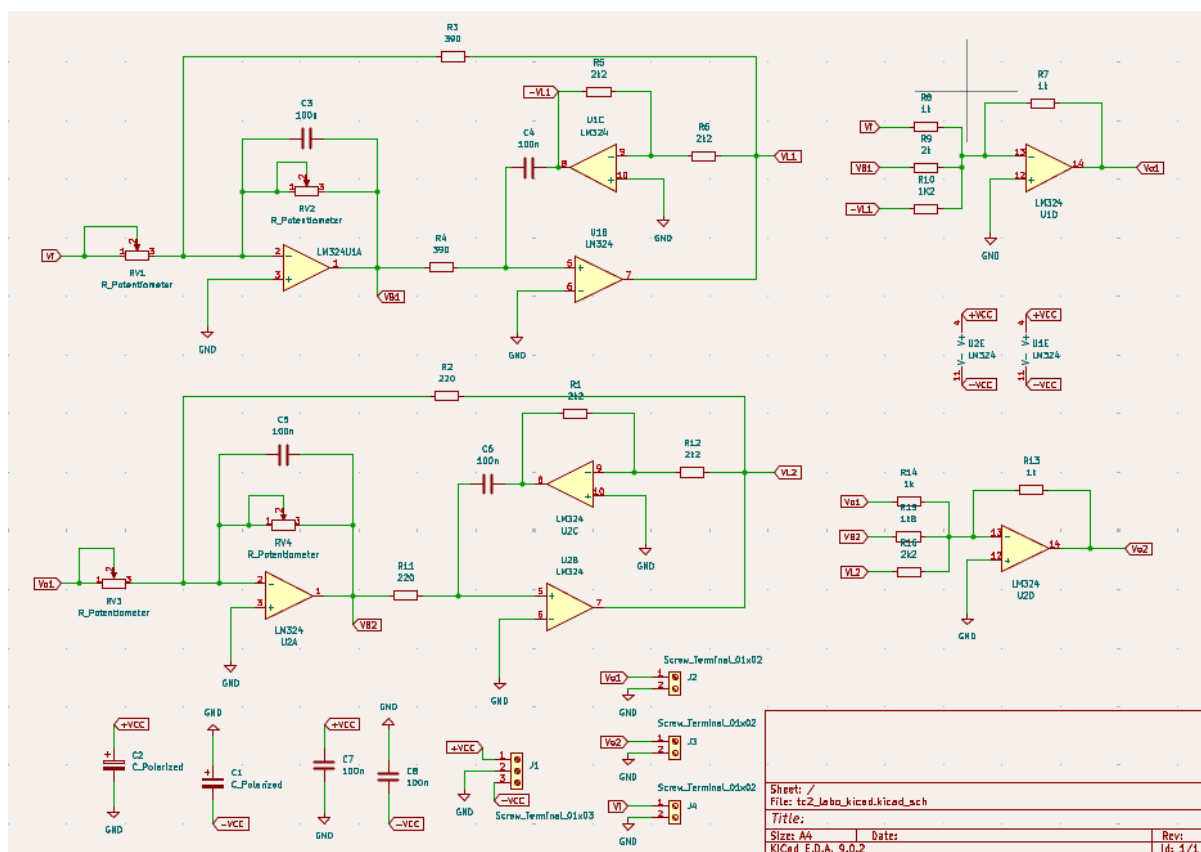
In Figure 20, it is possible to observe the first version of the connection made within the program for our Ackerberg-Mossberg band-removal filter.



Final schematic in “Altium Designer”.

In Figure 21 we present the second version of the schematic, changing the order of the stages -although we know that the result will be the same- and the value of the resistor that connects VB1 of $1k8\Omega$ a $2k\Omega$. This slightly improves the filter selection for the band that needs to be removed.

Gráfico 20



Final schematic in "KiCad".

In both cases, we added measurement terminals to the components we deemed necessary using pins, in addition to ground and power connections. We also added capacitors of $10\mu F$ electrolytic and another of $100nF$ in the power supplies to reduce high-frequency noise from the source.

4.2 PCB

Figure 22 shows the designed board with its connections for the first version.

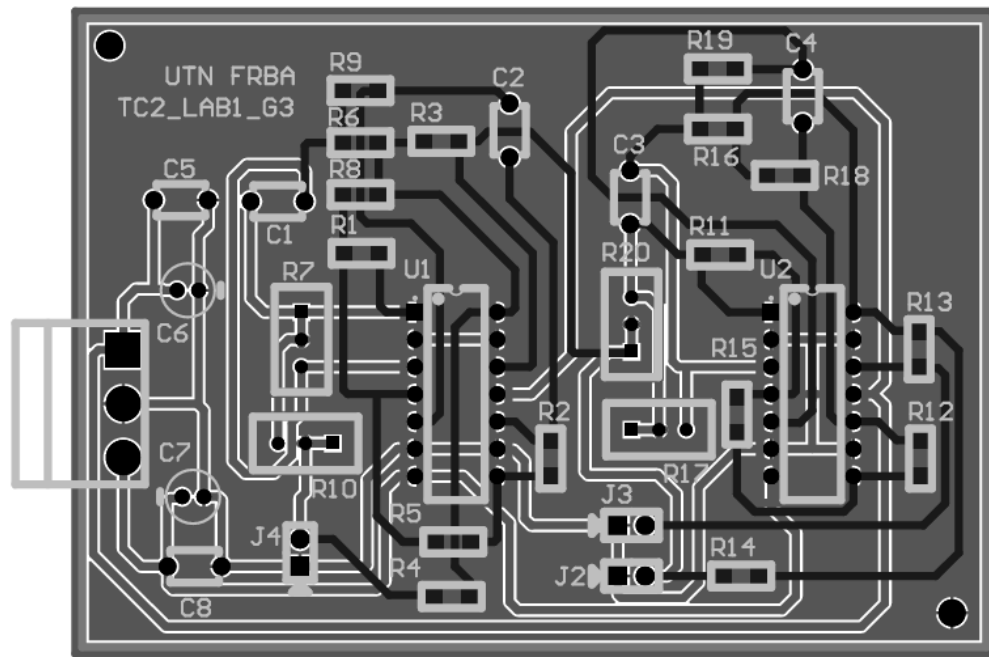


Gráfico 22

PCB final in "Altium Designer".

While in graph 23 we present the second version.

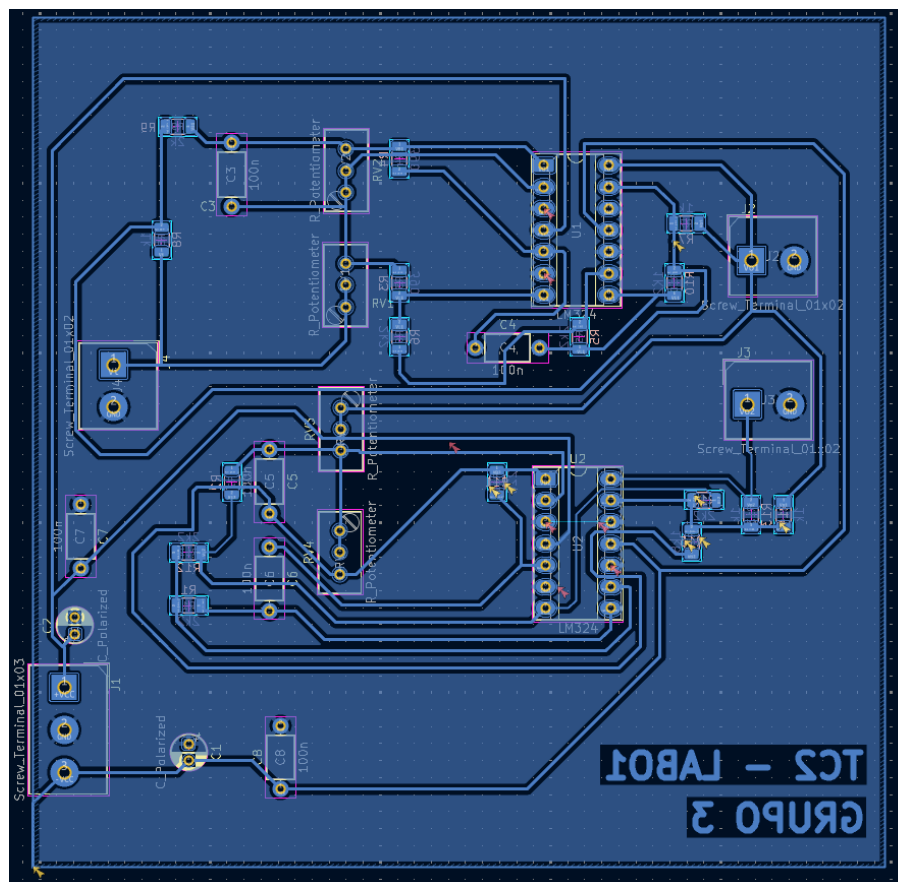


Gráfico 23

Final PCB in "KiCad".

5. MEASUREMENTS

5.1 Setup

To begin the measurements in both cases, a signal generator was used.(*graph 24*) to connect it to the system's input terminal. Then, channel 1 of the oscilloscope was connected to the circuit's input pins, and finally, channel 2 to its output.(*chart 25*).



Gráfico 24

Signal generator for system input.

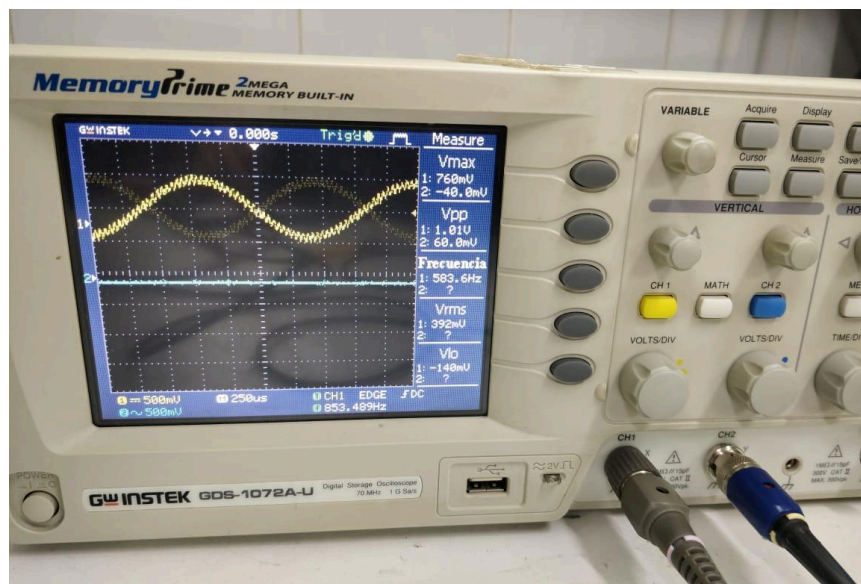


Gráfico 25

Oscilloscope used in practice.

We use a split source of $\pm 9V$ for feeding, as can be seen in graph 26.



Gráfico 26

Powering the system with a split power supply.

In Figures 27 and 28, we present both physical plates made by the team for the development of the practice.

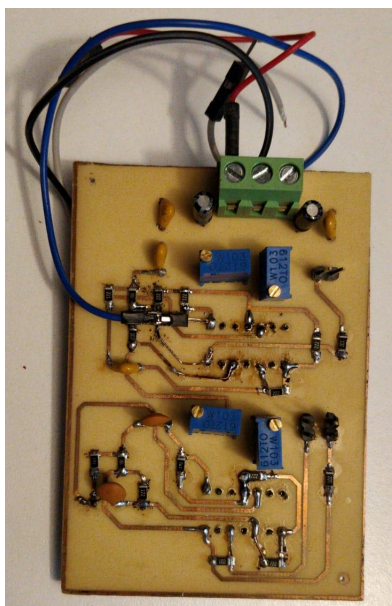


Gráfico 26

First version of the PCB board.

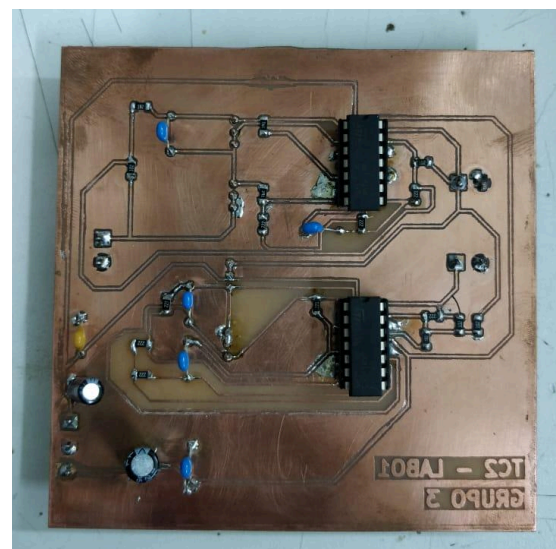


Gráfico 27

Second version of the PCB board.

We verified that the output signal from the generator was the same at the system input and began measuring from V_{o2} with a manual frequency sweep.

Despite several connection attempts and verifying that the components were correct, the first board did not respond as expected. The startup signal appeared slightly distorted at the input terminal, but once we progressed through the circuit, the

readings were quite peculiar. At low frequencies of approximately $500 - 1\text{kHz}$ (graph 28) the measurements both in V_{o1} as in V_{o2} They seemed to have higher frequencies than 1MHz (graph 29), similar to a signal modulation effect.

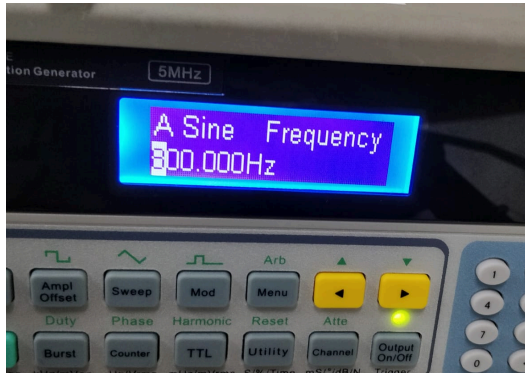


Gráfico 28

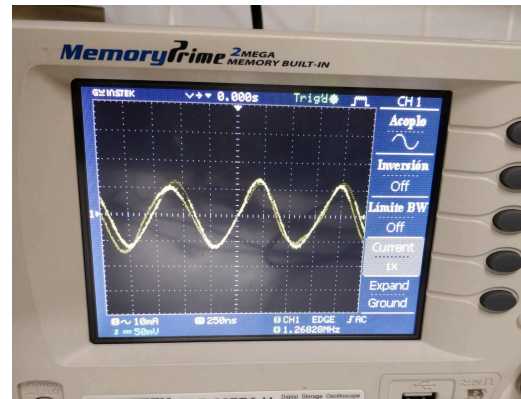


Gráfico 29

Signal generated and signal read at the output pins of the PCB board

Since the initial plate had many physical modifications made during the measurement in class, along with some alterations that we made in subsequent visits to the open laboratory; the team decided to design the second plate, focusing on maintaining the number of layers necessary to one.

The soldering was much simpler; we used terminal blocks instead of pins for the measurement points of interest, and the trimmers became more accessible for configuration. During the process, the value of one of the resistors was changed. (commercial value $1k8\Omega$ a $2k\Omega$) The first stage connects to the fourth amplifier to improve accuracy. Additionally, due to footprint availability in KiCad, we changed the operational amplifier model. (de $LM324$ a $TL084$) Once completed, the names of the power and reading points were added with a marker.

In Figure 30, we are using the intermediate terminals that we added to the design to be able to measure at the output $V_{o1} - V_{o2}$ and the entrance W_e .

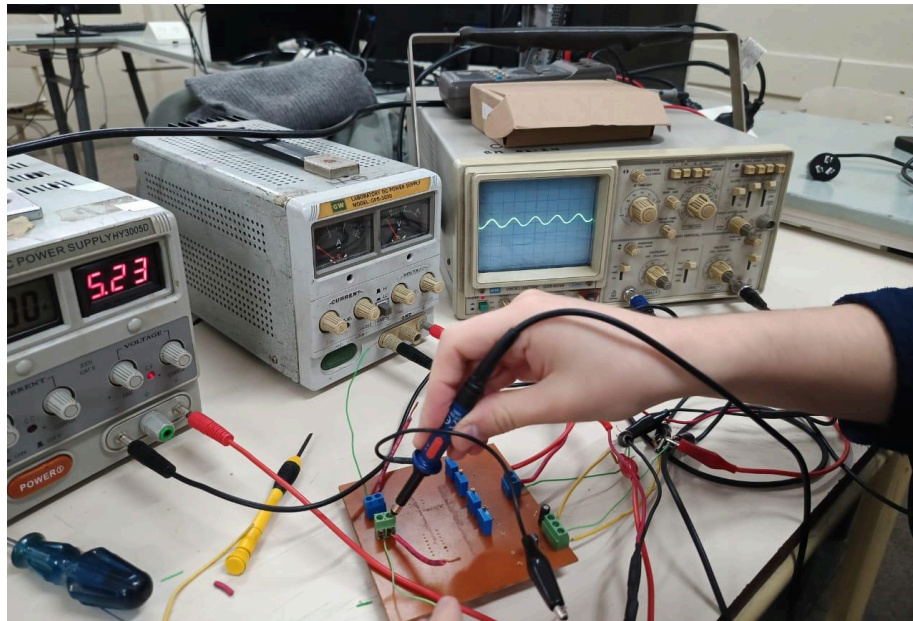


Gráfico 30

Initial measurements on the second version of the board.

The initial measurements were similar to those we saw on the first board, a signal modulation effect that changed the operating frequency of 1kHz a 10kHz . Therefore, we checked the schematic we had created and the PCB connections and realized there was an error in the labels connecting the second stage to the fourth amplifier. Once we corrected it in the file, we proceeded to cut the incorrect trace, wire the correct one, and double-check the connections.

It was decided on a sinusoidal input signal of variable frequency, with an amplitude of $2V_{pp}$; in graph 31 we repeated the measurements using the terminal blocks.

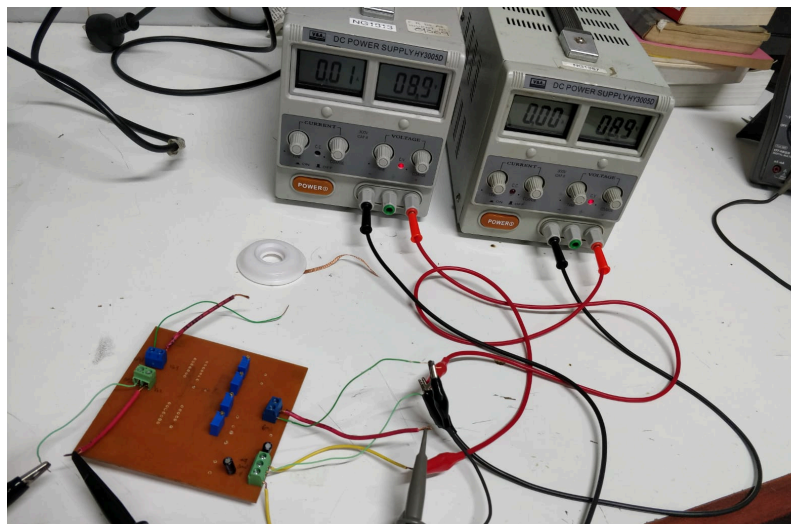


Gráfico 31

Second measurements on the second version of the plate.

However, it could be observed that the filter's response was limited to approximately 150mV. The voltage varied as we performed the frequency sweep, but it never exceeded that value, functioning as a low-pass filter. Again, we referred to our PCB model and the digitally created schematic to compare the connections. The source of this error turned out to be an SMD resistor that was making poor contact and interrupting the signal path.

Once we verified the contact of the remaining components, we attempted to measure the system's response again. What we observed was that the filter slightly modified the operating frequency, increasing it by approximately between 500 – 1kHz in each measurement.

Despite the aforementioned problems, it was decided to continue with the corresponding measurements and perform a subsequent analysis, focusing on the possible sources of error and uncertainty. The results obtained with the second plate are shown in Table 1.

N°	FrequencyIN [Hz]	FrequencyOUT [Hz]	Vout [V]	Phase shift [s]	Gain [dB]	Phase shift [rad]
1	200	1033	2.0000	0.000000	0.000000	0.000000
2	600	1576	2.0000	0.000020	0.000000	0.075398
3	1000	1754	2.0000	0.000052	0.000000	0.326726
4	1300	2155	2.0000	0.000056	0.000000	0.457416
5	1500	2376	2.0000	0.000068	0.000000	0.640885
6	1800	2406	2.0000	0.000068	0.000000	0.769062
7	2000	2715	2.0000	0.000068	0.000000	0.854513
8	2300	3082	1.9456	0.000071	-0.239296	1.026044
9	2600	3209	1.9120	0.000085	-0.390443	1.388584
10	2800	3761	1.7582	0.000086	-1.119036	1.512991
11	3000	3838	1.6089	0.000089	-1.889811	1.677610
12	3200	4086	1.3956	0.000090	-3.125353	1.809557
13	3500	4475	0.9476	0.000090	-6.487732	1.979203

14	4000	4938	0.3291	0.000082	-15.672911	2.060885
15	4500	5312	0.1097	0.000065	-25.215336	1.837831
16	4700	5351	0.0722	0.000056	-28.846419	1.653734
17	5000	5635	0.0740	0.000049	-28.629261	1.539380
18	5500	6272	0.0681	0.000042	-29.352047	1.451415
19	5800	6601	0.0470	0.000049	-32.567779	1.785681
20	6000	6952	0.0436	0.000064	-33.229944	2.412743
21	6500	7299	0.1796	0.000081	-20.932504	3.308097
22	7000	7784	0.4395	0.000082	-13.161300	3.606548
23	7500	8223	0.7555	0.000082	-8.455272	3.864159
24	8000	8889	1.1100	0.000083	-5.114140	4.172035
25	8500	9057	1.4000	0.000083	-3.098039	4.432787
26	9000	9767	1.7174	0.000082	-1.322756	4.636990
27	9600	10299	1.9401	0.000080	-0.263992	4.825486
28	10000	10657	2.0000	0.000079	0.000000	4.963716
29	12000	12711	2.0000	0.000071	0.000000	5.353274
30	15000	15791	2.0000	0.000060	0.000000	5.654867

Tabla 1

5.2 Gain and phase

It is possible to observe in graph 32 the graphical representation of the filter module response according to the measurements taken.

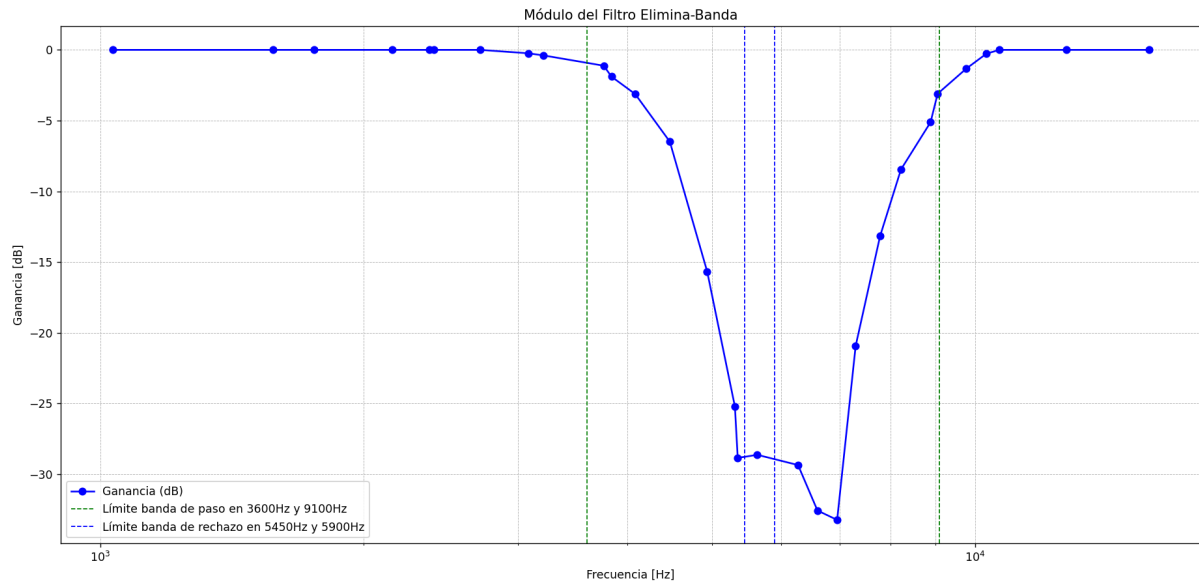


Gráfico 32

Graph of the band-stop filter module in dB.

On the other hand, in graph 33 we present the equivalent graph of the phase.

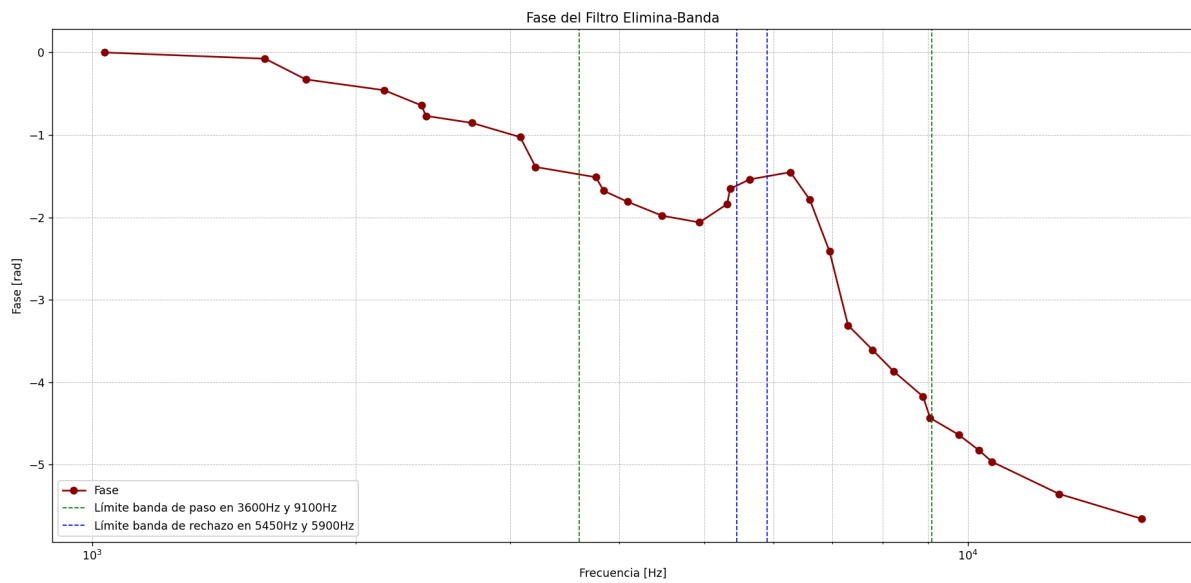


Gráfico 33

Graph of the band-removal filter phase in radians.

6. CONCLUSIONS

6.1 Design and simulation

Thanks to the exercises we did during the course and the examples provided in the book "Schaumann," we were able to create a design that met the required specifications without any problems. The only detail to consider is the chosen order; having opted for a design of $n = 2$ rather than $n = 3$. The filter exhibited lower selectivity, which affected the clarity of the frequency cutoff. While this decision was made to simplify the design, it's important to note that a higher order could have resulted in a more precise response.

6.2 PCB

The main problem arose from a lack of attention to detail when designing the first version of the board. Because we fabricated the double-sided board ourselves and didn't properly mirror the integrated circuit connections, we were forced to use wires and solder the sockets unstable.

These were taken into account for the second version, resulting in a single-layer board. However, we discovered that the KiCad labels are very sensitive if not stored correctly, and the output of the second circuit stage wasn't connected as it should have been. Therefore, a minor external modification was made to the board, and it was re-engineered. Furthermore, we were very careful when soldering the SMD resistors and other through-hole components, using a soldering iron with interchangeable tips for greater convenience.

6.3 Measurements

To conduct experiments like those in this lab, it is crucial to use certified and calibrated instruments, thereby reducing potential measurement uncertainty. Furthermore, the number of cables used should always be minimized, and if necessary, high-quality cables should be selected to avoid losses or unwanted noise.

Thanks to our multiple visits to the lab for modifications and new measurements, we were able to design and evaluate two physical versions of the printed circuit board (PCB). However, we were unable to achieve the exact results we expected from the simulations performed in LTSpice or with the theory covered in class.

One of the most significant differences was the phase shift between the input and output, as it did not exactly match the expected behavior for a second-order filter. This can be explained by slight variations in the actual values of the components

used—plus their tolerances—as well as by the presence of parasitic impedances on the PCB traces and the physical layout of the circuit.

Finally, we observed that the circuit exhibited small shifts in operating frequency, with increases on the order of 500Hz a 1kHz . This caused the expected cutoff frequency to be completely shifted. This could be due to variations in the resistor and capacitor values, along with the influence of external connections made to correct errors in the second version of the PCB.

6.4 Comments

We were able to characterize the overall operation of the band-stop filter, identify the most relevant sources of error, and analyze their effect on system performance, all despite the obstacles we faced. Thanks to this project, we gained a deeper understanding of technical and practical terms, recognizing the relationship between design, simulation, implementation, and measurement in a real laboratory environment.